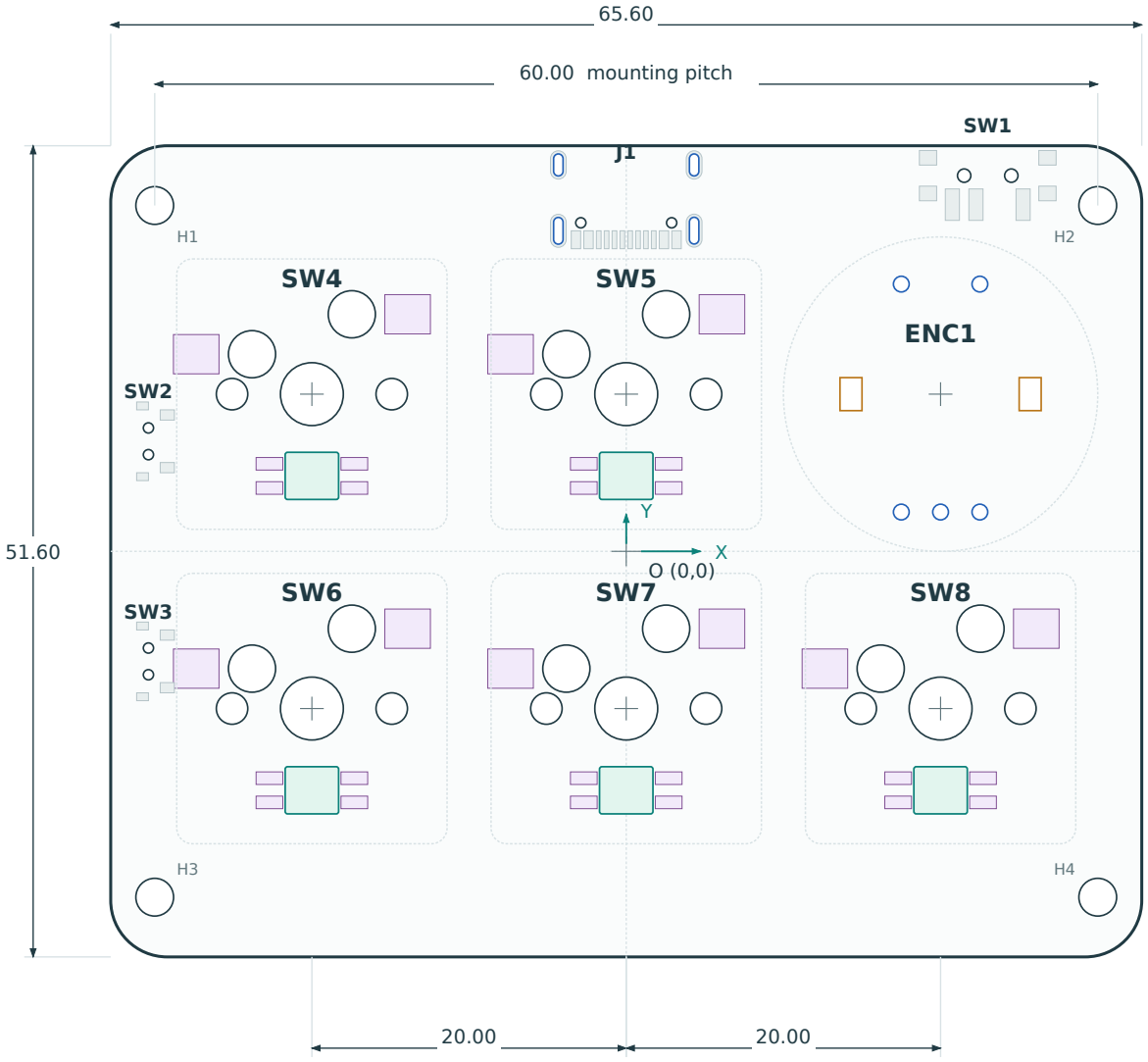
CH592 / PCB mechanical layout

Board 65.6 × 51.6 × 1.6 · R3.6 corners · 20 mm key pitch · Print06 structural proposal



FRONT / USER SIDE · dashed = flange clearance / knob envelope

DATUM COORDINATES

Ref.	X	Y
SW4	-20.00	+10.00
SW5	+0.00	+10.00
SW6	-20.00	-10.00
SW7	+0.00	-10.00
SW8	+20.00	-10.00
ENC1	+20.00	+10.00
J1	+0.00	+22.20
SW1	+23.00	+23.90
SW2	-30.40	+7.00
SW3	-30.40	-7.00

H1 / H2: X = ±30, Y = +22

H3 / H4: X = ±30, Y = -22

4 × Ø2.40 NPTH for M2 screws

- PCB mechanical / NPTH hole
- PCB signal / PTH hole or slot
- B.Cu pad (top view; no extra mirror)
- RGB through-board light window
- ENC1 shell slots: plating TBD

Board plane dimensions before 6° tilt.

Lower-left bounding-box origin:

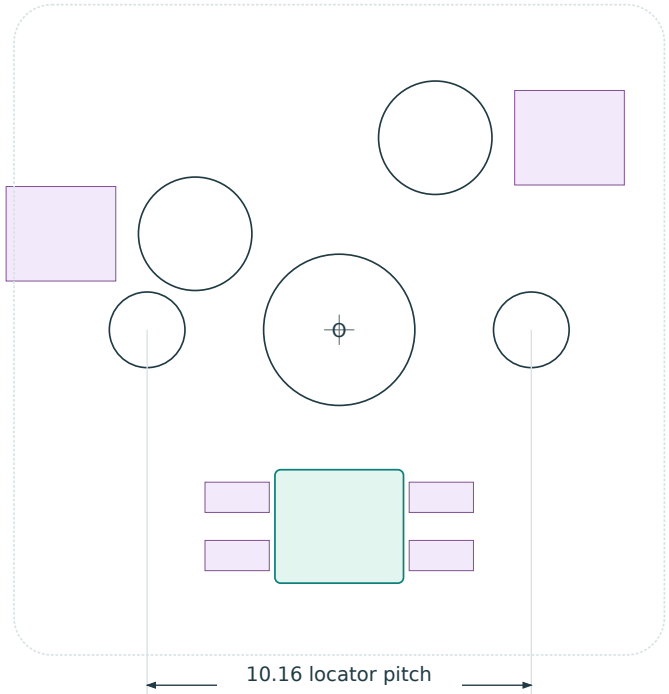
XII = X + 32.8; YII = Y + 25.8

CH592 / Hot-swap switch and encoder details

All geometry in PCB TOP VIEW. Purple lands are B.Cu. Switch and encoder datums are their stem / shaft axes.

A SAMZO 28115181DM + C49352235

5 keys · switch pattern shown once · detail scale 5:1

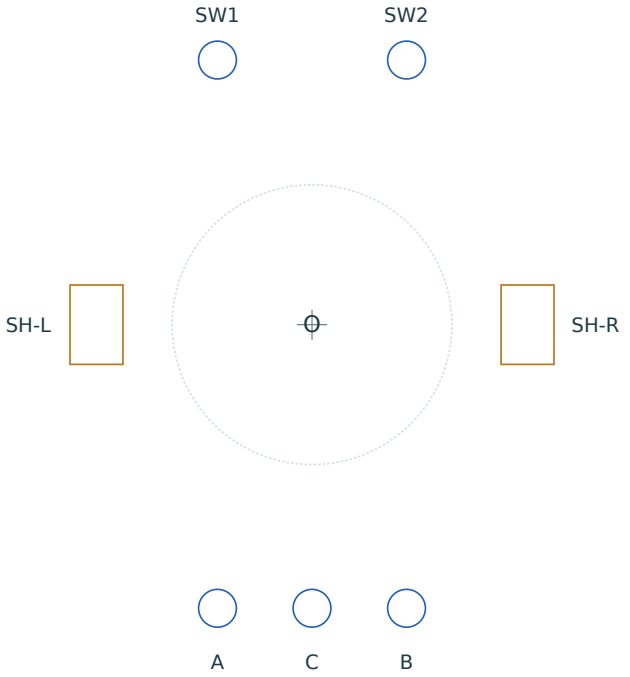


Feature	Local X	Local Y	Size / type
Stem locator	0.00	0.00	Ø4.00 NPTH*
Side locators	±5.08	0.00	2 × Ø2.00 NPTH*
Socket pin 1	-3.81	+2.54	Ø3.00 NPTH
Socket pin 2	+2.54	+5.08	Ø3.00 NPTH
B.Cu pads	-7.36 / +6.09	+2.54 / +5.08	2.90 × 2.50

* Existing-design clearances; verify SAMZO fit. No key Ø1.20 PTH remains.

B Bourns PEC11L-4115K-S0020 / ENC1

PROPOSED REPLACEMENT · user hardware change not confirmed · 5:1



Feature	Local X	Local Y	Drill / cut
A / C / B	-2.50 / 0.00 / +2.50	-7.50	Ø1.00 +0.20/-0
SW1 / SW2	±2.50	+7.00	Ø1.00 +0.20/-0
SH-L / SH-R	±5.70	0.00	1.40 × 2.10 slot
Lid shaft bore	0.00	0.00	Ø7.40 - lid only

Ø6 / 18 teeth shaft; tip at PCB +19.50; body 4.50 + L15.

No copper annuli specified; shell-slot plating TBD by PCB design.

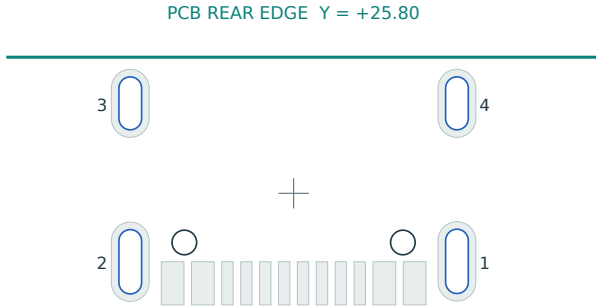
K-shaft LCSC availability unverified; order this exact suffix.

CH592 / USB, power and side service access

PCB datums in X/Y. Sidewall opening centers use app (x,y,z), before the whole enclosure 6° rotation.

C USB-C / J1

GT-USB-7010ANS · pad detail 5:1



PCB feature	Local XY / drill
Shell 1 / 2	±4.32,-1.80 / 0.60×1.70
Shell 3 / 4	±4.32,+2.38 / 0.60×1.40
Locators	±2.89,-1.30 / Ø0.65
Origin / angle	(0,+22.20) / 180°

CASE OPENING (not PCB cut)

Throat 9.60 × 3.80, R1.20

Outer mouth 10.00 × 4.20

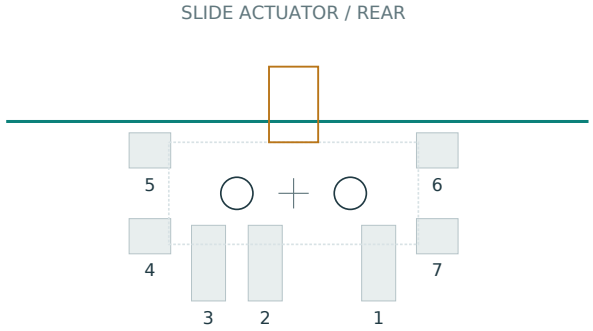
3D center: 0, 15.28, -28

Outer entry chamfer C0.20.

Copper nearest rear edge: 0.32 mm nominal.

D POWER / SW1

C963206 · MK-12C02-G020 · 5:1



Item	mm
Body center	(+23,+23.90)
Body / travel	6.60×2.70×1.40 / 1.50
2 locator holes	Ø0.85 at X = ±1.50
Signal lands	0.90×2.00
Anchor lands	1.10×0.93

LCSC / EasyEDA enlarged copper lands.

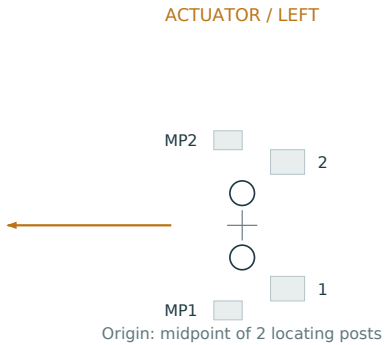
Body center differs from library header origin.

M2 SELF-TAPPING FASTENERS

Lid pilot Ø1.80, depth 4.90.

E RESET / BOOT

C2762970 · side-actuated · 5:1



Item	mm
SW2	(-30.40,+7.00)
SW3	(-30.40,-7.00)
Envelope	4.7×3.75×1.9
2 locator holes	Ø0.65 NPTH (proposal)

LEFT WALL ACCESS (2 ports)

Each Ø1.60; axis app +X.

Opening centers app (x,y,z):

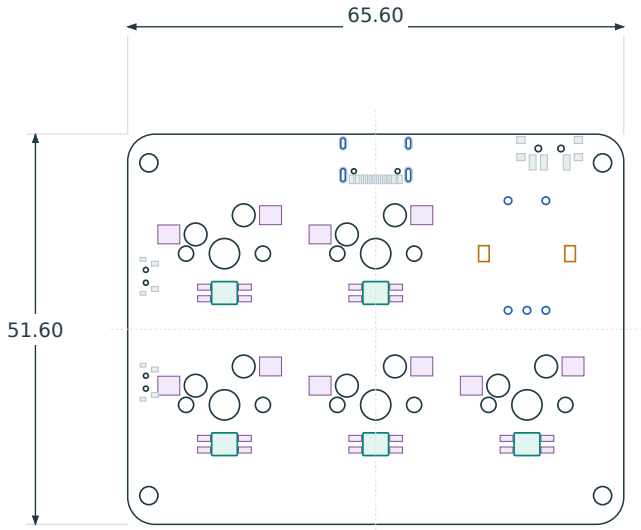
-35, 14.55, -7

-35, 14.55, 7

Pad size / holes: proposal; verify footprint.

CH592 / 1:1 alignment reference

Print at Actual size / 100%. Disable Fit to page. Check the 50 mm scale bar before comparing against hardware.



FRONT / USER



50.00 mm calibration

This page is an alignment template, not a drill program.

Sidewall access is excluded from this PCB template.

M2 mounting holes: Ø2.40, non-plated.

RELEASE NOTES

01 Datums

All coordinates are in the PCB plane.

X increases right; Y increases toward USB.

App transform: appx = X, appz = -Y.

02 Hot-swap pattern

C49352235: 2 × Ø3.00 NPTH per switch.

Two 2.90 × 2.50 B.Cu lands per socket.

SAMZO stem locator clearances need sample fit.

03 RGB / service access

D2-D6 are on B.Cu, mirrored in X in top view.

Confirm pin 1 and chain direction before routing.

SW2 / SW3 access ports are in the left sidewall.

04 Enclosure and PCB release

M2 self-tappers engage resin pilot holes in the lid.

PCB holes remain Ø2.40 NPTH clearance holes.

ENC1 Bourns replacement is proposed, not confirmed.

Files: 5 SVG sheets, PDF, coordinates CSV,

PCB / retention DXF, JSON and Chinese notes.

CH592 / Switch retention and socket stack

Integral lid ledges retain the switch clips. All dimensions are local, before enclosure tilt; plate features are CASE_ONLY.

F SWITCH LOCATING OPENINGS

Top view 2:1 · centers shared with PCB · orange = retaining aperture



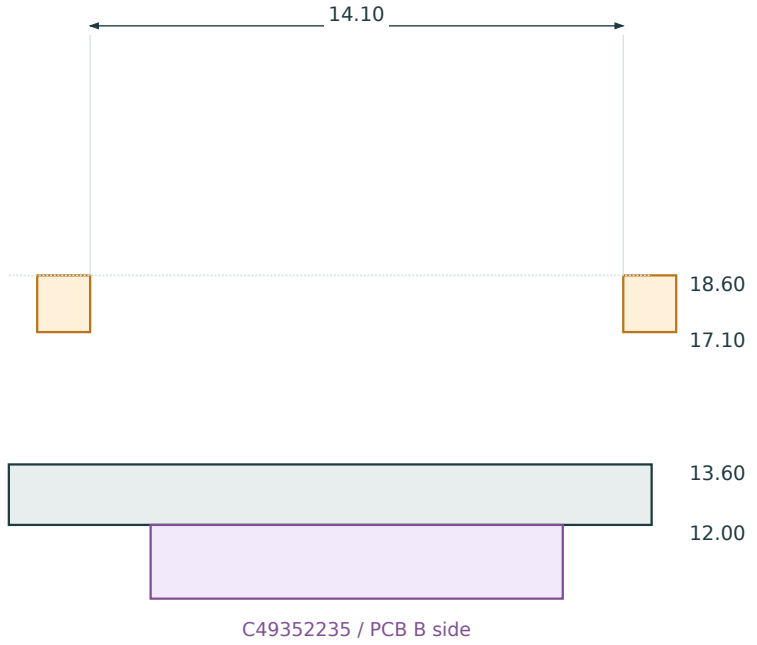
5 × 14.10 × 14.10, R0.15

Dashed flange clearance: 17.20 × 17.20, R1.00.

Fit coupons: 14.00 / 14.10 / 14.20 square openings.

G LOCAL VERTICAL STACK

Section through switch center · 5:1 · app Y heights in mm



Retention seat: PCB top + 5.00 = app 18.60.

Effective ledge thickness: 1.50 mm.

Socket metal: 1.85 ±0.10; use 1.95 below PCB.

3.05 total socket height includes PCB insertion.